

Trainings ▾

Initial Check

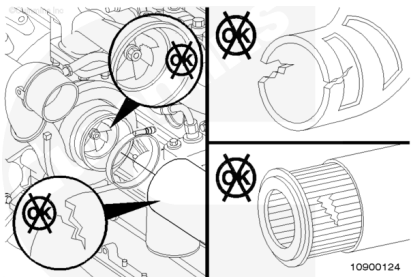
Remove the intake pipe from the turbocharger. See the Remove step in this procedure.

Inspect the turbocharger compressor impeller blades for damage.

Replace the turbocharger if damage is found. See the Remove and Install steps in this procedure.

If the compressor impeller is damaged, inspect the intake piping and filter element for damage.

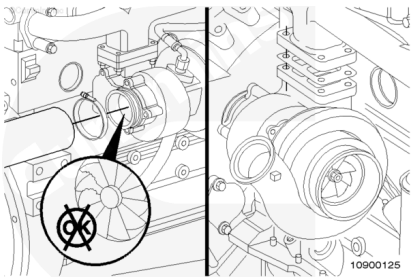
Repair any damage before operating the engine.



Remove the exhaust pipe from the turbocharger. See the Remove step in this procedure.

Inspect the turbine wheel for damage.

Replace the turbocharger if damage is found. See the Remove and Install steps in this procedure.

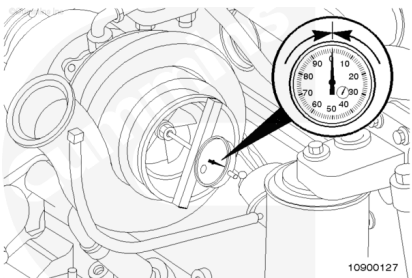


Axial Clearance Check

Use dial depth gauge, Part Number ST-537.

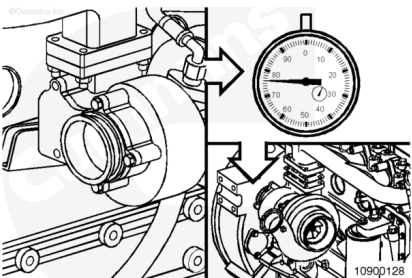
Push the rotor assembly away from the gauge.

Set the gauge on zero.



Push the rotor assembly toward the gauge and record the reading.

Axial Clearance		
mm		in
0.025	MIN	0.001
0.127	MAX	0.005



Replace the turbocharger if the clearance does **not** meet the specifications. See the Remove and Install steps in this procedure.

Inspect the compressor wheel for signs of rubbing against the compressor cover.

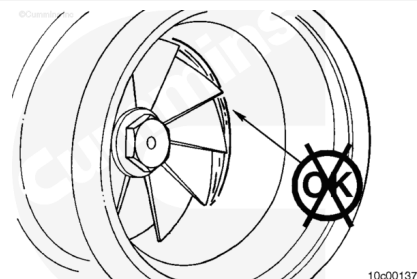
Replace the turbocharger if rubbing evidence is seen.

Check the radial movement of the turbocharger wheels and shaft.

Use light finger pressure to push the compressor wheel toward the side of the compressor housing.

Repeat the procedure for the turbine wheel and housing.

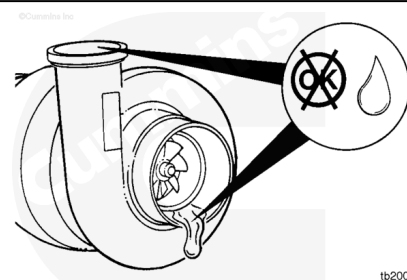
Replace the turbocharger if either wheel contacts the housing.



Leak Check

Inspect the compressor intake and discharge for oil.

If oil is present in the compressor intake as well as in the discharge, check upstream in the turbocharger for the source of the oil.

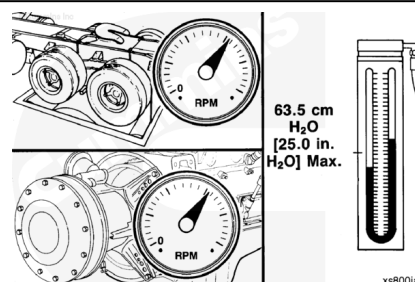


If oil is present **only** in the discharge side, install the air intake and charge air cooler piping.

Check for intake restriction. Refer to Procedure 010-031 (</qs3/pubsys2/xml/en/procedures/100/100-010-031.html>).

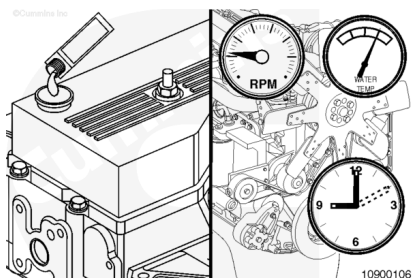
If excessive intake restriction is **not** found, replace the turbocharger. See the Remove and Install steps in this procedure.

Note : If the engine experiences a turbocharger malfunction or any other occasion where oil is put into the charge air system, the charge air system must be inspected and cleaned. Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/100/100-010-027.html>).



Add 1 unit of fluorescent tracer, Part Number 3376891, to each 38 liters [10.0 gal] of engine lubricating oil.

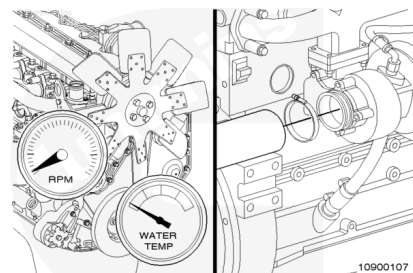
Operate the engine at low idle for 10 minutes.



Shut off the engine.

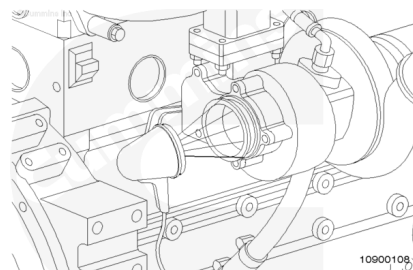
Allow the turbocharger to cool.

Remove the exhaust pipe from the turbine housing. See the remove step in this procedure.



Use a high-intensity black light, Part Number 3163339, to inspect the turbine outlet for leaks.

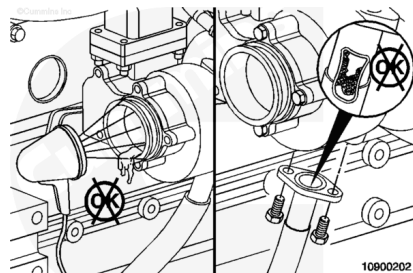
A yellow glow indicates an oil leak. A dark blue glow indicates fuel in the oil.



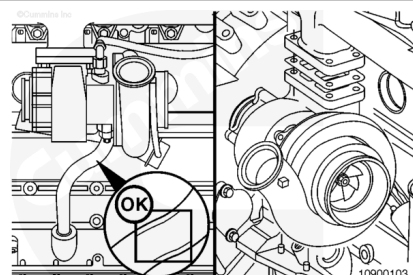
If oil is found on the turbine housing, remove the oil drain line and check for restrictions. Refer to Procedure 010-045 (</qs3/pubsys2/xml/en/procedures/100/100-010-045.html>).

Clear any restrictions found.

Install the turbocharger oil drain line and new o-ring seals into the engine block. Refer to Procedure 010-045 (</qs3/pubsys2/xml/en/procedures/100/100-010-045.html>).



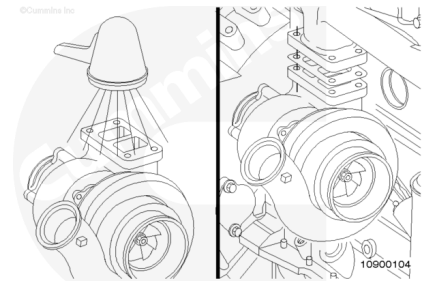
If the oil drain line was **not** restricted, remove the turbocharger. See the Remove step in this procedure.



Use a high-intensity black light, Part Number 3163339, to inspect the turbine inlet for leaks.

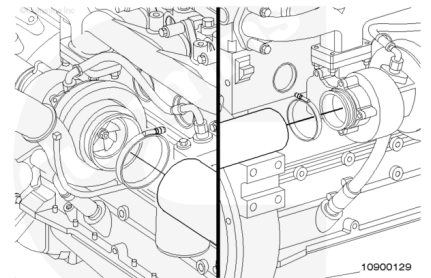
A yellow glow indicates an oil leak from the engine.

If a yellow glow is **not** seen in the turbine inlet, replace the turbocharger. See the Remove and Install steps in this procedure.



Install the exhaust pipe to the turbocharger turbine outlet and tighten the clamp. See the Install step in this procedure.

Install the intake pipe to the turbocharger compressor inlet and tighten the clamp. See the Install step in this procedure.



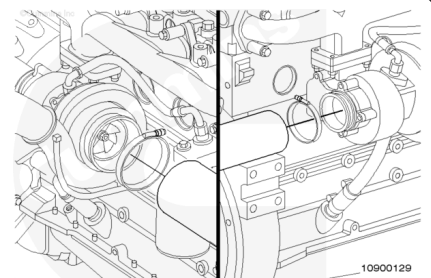
Preparatory Steps

- Remove the air crossover if equipped. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/100/100-010-019.html>).
- Disconnect the charge air cooler hose. See equipment manufacturer service information.
- Remove the oil supply line from the turbocharger. Refer to Procedure 010-046 (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>).
- Remove the oil drain line from the turbocharger. Refer to Procedure 010-045 (</qs3/pubsys2/xml/en/procedures/100/100-010-045.html>).
- Remove the intake manifold pressure supply line from the boost capsule, if equipped.

Remove

⚠ WARNING ⚠

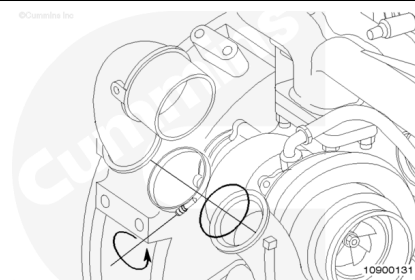
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the exhaust piping.

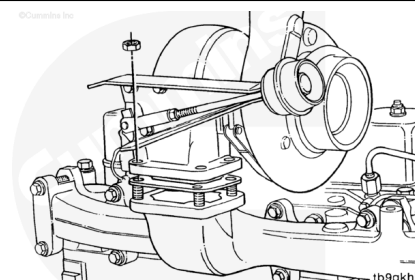
Remove the turbocharger compressor air inlet pipe.

Remove the turbocharger compressor outlet elbow, V-band clamp, and o-ring from the turbocharger compressor outlet.

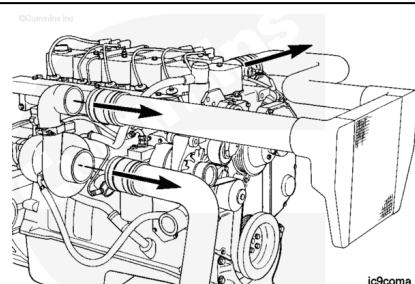


Remove the four turbocharger mounting nuts.

Remove the turbocharger and gasket.



Disconnect the charge air cooler piping. See equipment manufacturer service information.



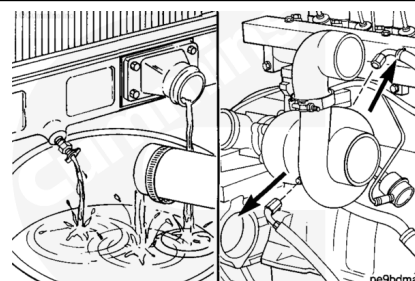
Disconnect the exhaust piping. See equipment manufacturer service information.

⚠ WARNING ⚠

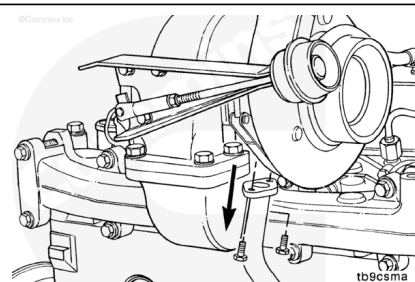
Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

Drain the coolant; Refer to Procedure 008-018
(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html).

Disconnect the turbocharger coolant lines.

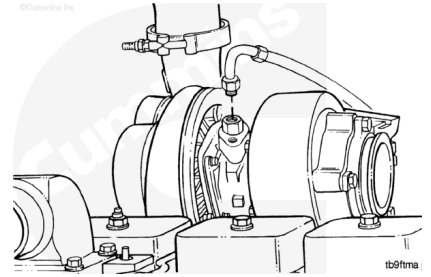


Remove the capscrews from the turbocharger oil drain tube.

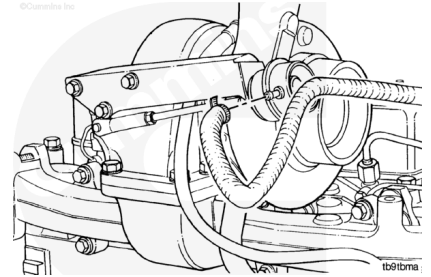


Disconnect the turbocharger oil supply line.

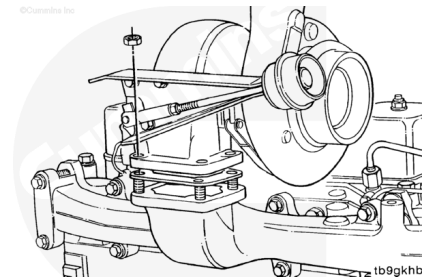
Note : C8.3G Engines use a copper washer in both ends of the turbocharger lubricating oil supply line. The washer should be replaced whenever the line is removed.



Remove the intake manifold pressure supply line from the boost capsule.



Remove the mounting nuts, turbocharger, and gasket.



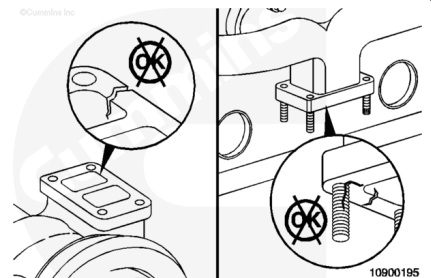
Clean and Inspect for Reuse

Clean the turbocharger and exhaust manifold gasket surfaces.

Inspect the turbocharger and exhaust manifold gasket surfaces, and mounting studs for cracks and damage.

Replace the turbocharger if any cracks are found on the mounting flange surface. See the Remove and Install steps in this procedure.

Replace the exhaust manifold if any cracks are found on the mounting flange surface. Refer to Procedure 011-007 (</qs3/pubsys2/xml/en/procedures/41/41-011-007.html>).



When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

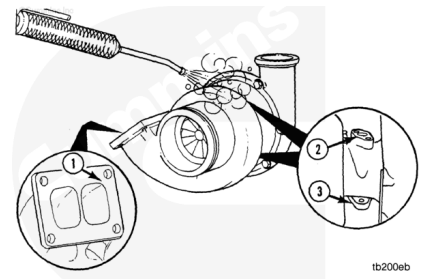
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Remove all carbon deposits and gasket material from surfaces (1, 2, and 3).

Use solvent or steam to clean the exterior of the turbocharger.

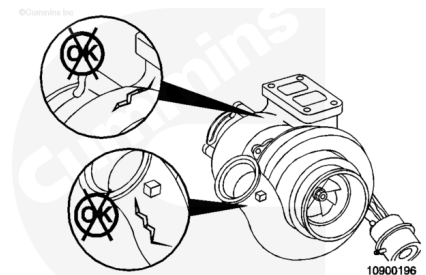
Dry with compressed air.



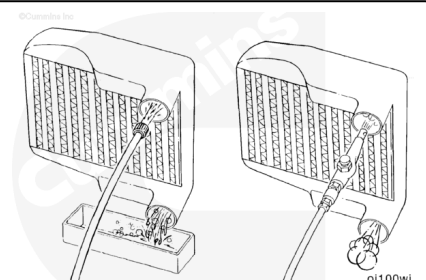
Inspect the turbine and compressor housings.

If cracks that go all the way through the outer walls are found, the turbocharger **must** be replaced.

Note : A charge air cooler malfunction can cause progressive damage to the turbine housing. If the turbine housing is damaged, check the charge air cooler. Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/100/100-010-027.html>).



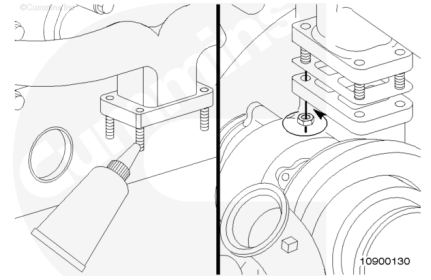
Note : If the engine experiences a turbocharger malfunction or any other occasion where oil or debris is put into the charge air system, the charge air system **must** be inspected and cleaned. Refer to Procedure 010-027 (</qs3/pubsys2/xml/en/procedures/100/100-010-027.html>).



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Apply a film of high-temperature anti-seize compound to the turbocharger mounting studs.

Use a new gasket and install the turbocharger.

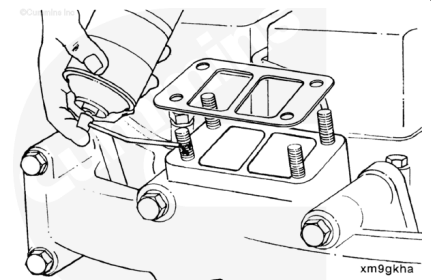
Install and tighten the four mounting nuts.

Note : The torque values given have been established using anti-seize compound as a lubricant.

Torque Value: 45 n•m [33 ft-lb]

Install a new turbocharger gasket.

Apply anti-seize compound to the mounting studs.



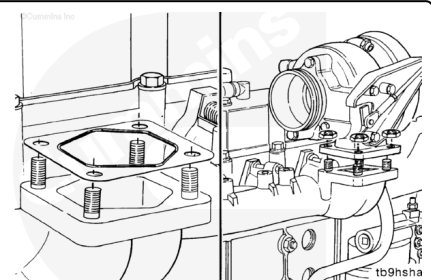
Install the turbocharger.

Install and tighten the turbocharger mounting nuts.

Torque Value: 45 n•m [33 ft-lb]

Use a new gasket, and connect the oil drain tube.

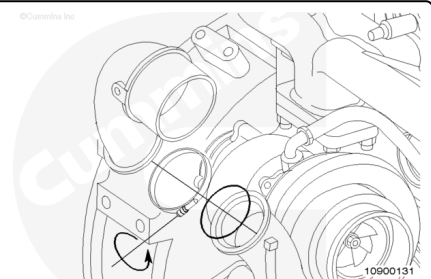
Torque Value: 24 n•m [18 ft-lb]



Install the turbocharger compressor outlet elbow, V-band clamp, and new o-ring seal on the turbocharger compressor discharge outlet.

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

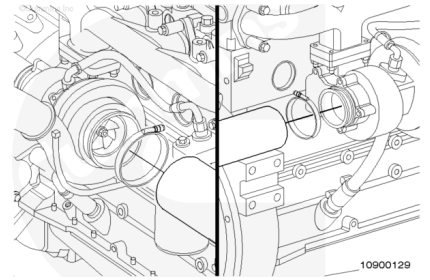


Install the intake pipe to the turbocharger compressor inlet and tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

Install the exhaust pipe to the turbocharger turbine outlet and tighten the clamp.

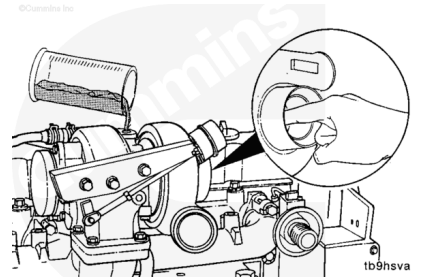
Torque Value: 8 n•m [71 in-lb]



Note : The turbocharger must be prelubricated before start-up.

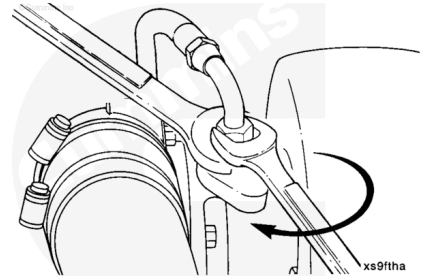
Pour 50 to 60 cc [2 to 3 fl oz] of clean engine oil into the oil supply fitting.

Rotate the turbine wheel to allow the oil to enter the bearing housing.

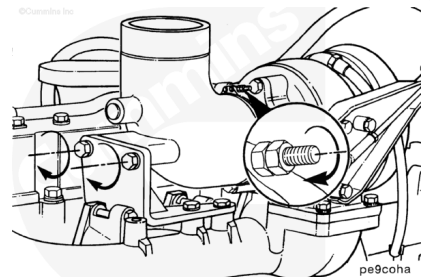


Connect and tighten the turbocharger oil supply line.

Torque Value: 35 n•m [26 ft-lb]



If required, loosen the compressor housing V-band clamp, and position the housing to align it with the turbocharger air outlet tube.

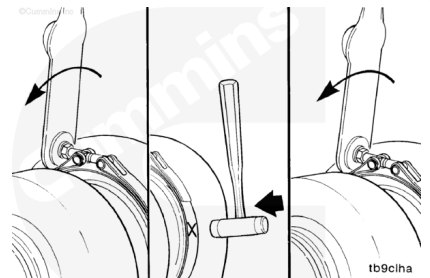


Tighten the band clamp.

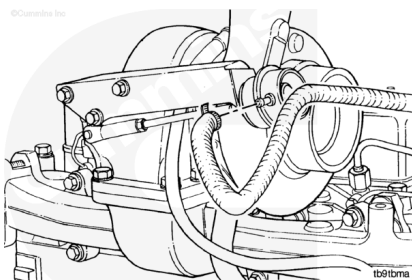
Torque Value: 8.5 n•m [75 in-lb]

Tap around the clamp with a plastic hammer and tighten again.

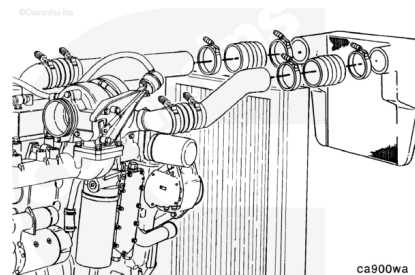
Torque Value: 8.5 n•m [75 in-lb]



Install the boost control capsule actuator hose.



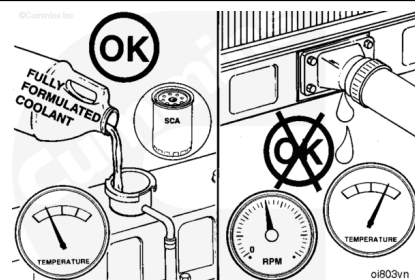
Install the charge air cooler hoses. See equipment manufacturer service information.



Install the air inlet and exhaust piping. See equipment manufacturer service information.

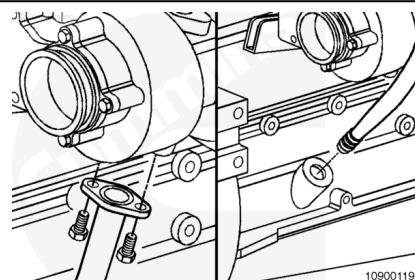
Fill the cooling system; Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).

Operate the engine and check for leaks.



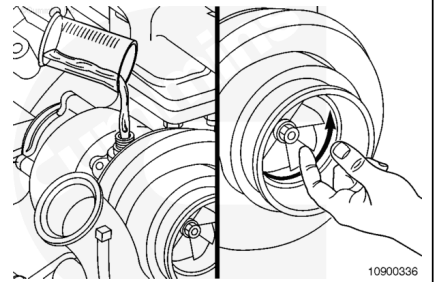
Prime

Install the turbocharger oil drain line. Refer to Procedure 010-045 (</qs3/pubsys2/xml/en/procedures/100/100-010-045.html>).



Lubricate the bearings by pouring 59 to 89 ml [2 to 3 oz] of clean 15W40 engine oil into the turbocharger oil supply line fitting. Rotate the turbine wheel to allow oil to enter the bearing housing.

Install the turbocharger oil supply line. Refer to Procedure 010-046 (</qs3/pubsys2/xml/en/procedures/100/100-010-046.html>).



Finishing Steps

- Install the intake manifold pressure supply line to the boost capsule, if equipped.
- Connect the charge air cooler hose. See equipment manufacturer service information.
- Install the air crossover, if equipped. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/100/100-010-019.html>).
- Operate the engine and check for leaks.